SAURAV CHOURASIA

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Experience

FOSSEE Summer Fellowship, IIT Bombay May 2026 – July 2026
Software Engineering Intern – eSim AI Assistant *Remote | GitHub*

- Redesigned the eSim AI assistant pipeline to extract circuit and simulation facts deterministically (netlist structure, simulation setup) before LLM inference, so the model explains verified facts instead of inferring them.
- Engineered a rule-based NgSpice error analysis engine that detects, ranks, and suppresses cascading errors to identify root causes before generating user-facing explanations.
- Developed an offline circuit analysis workflow that generates structured netlist summaries and error reports without requiring an LLM, keeping core analysis available on systems without local AI.
- Integrated the analysis workflow into eSim, enabling netlist analysis from the Project Explorer and connecting simulation failures directly to the AI assistant.

Projects

Snip & Ask | *Chrome Extension · OCR · Cloudflare Workers · D1 · LLM APIs* GitHub | Chrome Web Store

- Published a Chrome extension that lets users snip any part of their screen and ask AI about it, reaching **2.5K+ installs and 1300+ users** on the Chrome Web Store.
- Developed a self-hosted Cloudflare Worker with D1-backed rate limiting for Guest Mode, serving **55K+ API requests** without requiring users to configure API keys.
- Implemented automatic model routing that sends screen captures through OCR to text models and falls back to vision models only when visual context is required.
- Added multi-provider support with **BYOK and Compare Mode**, allowing users to connect their own AI providers and compare responses across multiple models.

Modular RAG & Retrieval Benchmark | *Python · ChromaDB · BM25 · Cross-Encoder* GitHub

- Built a modular RAG pipeline from scratch with swappable ingestion, chunking (6 strategies), retrieval (4 strategies), and generation components (cited-span, abstention, Chain-of-Verification), supporting Gemini and local Ollama backends.
- Wrote Recall@k, Precision@k, MRR, and NDCG@k from scratch and benchmarked semantic, BM25-hybrid, and cross-encoder reranked retrieval on BEIR SciFact, ArguAna, and NFCorpus.
- Found hybrid + reranking scored highest on SciFact (0.665 NDCG@5) and NFCorpus (0.386), while plain semantic won on ArguAna (0.315), so the best strategy depends on the corpus.
- Analyzed the quality/latency trade-off: semantic + reranking came within 0.006 NDCG@5 of hybrid + reranking on SciFact at 0.3s vs up to 1.5s per query, and Python BM25 scoring slowed from 0.17s to 1.19s as the corpus grew from 3.6k to 8.7k documents.

LLM ChatBot Backend | *Python · FastAPI · PostgreSQL · SQLAlchemy* GitHub | Demo

- Designed and deployed a multi-user chatbot backend with FastAPI and PostgreSQL supporting authenticated conversations, persistent chat history, and REST APIs.
- Implemented JWT authentication, SQLAlchemy ORM, Alembic migrations, and conversation-level authorization to enforce user data ownership.
- Streamed LLM responses to the client in real time using Server-Sent Events (SSE), and added Redis user caching and Docker-based deployment.

Technical Skills

Languages: Python, C++

AI / LLM: RAG, Agentic AI, LangChain, LangGraph

Backend: FastAPI, REST APIs, SQLAlchemy, Alembic, PostgreSQL

Tools & Infrastructure: Git, Cloudflare Workers, Cloudflare D1

Certifications

AWS Certified AI Practitioner View Certificate

Education

Vellore Institute of Technology May 2027
Bachelor of Technology in Computer Science and Engineering *CGPA: 8.48 / 10*